

Tableau Basics

Q1. Explain what Tableau is and why it is preferred over traditional spreadsheet tools for data analysis. Also mention any two industries where Tableau is commonly used and explain one use case.

Ans. Tableau is a data visualization and business intelligence tool used to analyze and present data. It is preferred over spreadsheets because it can handle large datasets, create interactive dashboards, and provide better visual analysis. Industries: Retail and Healthcare. Use case: Retail companies use Tableau to track sales performance and customer trends.

Q2. After connecting to the Sample – Superstore dataset:

a) List the main types of fields available (examples: time-based, geographical, numerical).

b) Identify any four Dimensions and any four Measures from the dataset.

Ans.

a) Main field types: Time-based (Order Date), Geographical (State, Region), Numerical (Sales, Profit).

b) Dimensions: Category, Sub-Category, Region, State.

Measures: Sales, Profit, Quantity, Discount.

Q3. Create a Bar Chart showing Total Sales by Category. Explain:

Which field goes to Rows

Which field goes to Columns

Which aggregation is applied to Sales

Ans.

To create a Bar Chart showing Total Sales by Category:

- Category → Rows
- Sales → Columns
- Aggregation applied to Sales → SUM(Sales)

This chart helps compare sales across categories.

Q4. What insight can be gained by comparing Sales and Profit together? Then explain how you would create a chart to analyze Profit by Sub-Category.

Ans. Comparing Sales and Profit helps identify whether high sales are generating good profits. To analyze Profit by Sub-Category, drag Sub-Category to Rows and Profit to Columns, then create a bar chart.

Q5. Create a Line Chart to show Sales Trend over Order Date.

Which date level would you choose (Year / Quarter / Month)?

Why are line charts suitable for time-based analysis?

Ans. I would choose Month as the date level because it shows trends clearly over time. Line charts are suitable for time-based analysis because they display changes and patterns across a timeline.

Q6. Apply a Basic Filter to display data only for:

Region = “West”

Category = “Technology”

Explain the steps and mention which shelf is used.

Ans.

Steps:

1. Drag Region to the Filters shelf and select West.
 2. Drag Category to the Filters shelf and select Technology.
- The Filters shelf is used to display only the required data.

Q7. Explain the business importance of geographical analysis.

Then describe how you would create a Map showing Sales by State using Sample – Superstore

Ans. Geographical analysis helps businesses understand performance across different locations. To create a map, drag State to the canvas and then drag Sales onto the map. Tableau automatically creates a map visualization.

Q8. What is the difference between a Worksheet and a Dashboard in Tableau? Why is a dashboard more useful for decision-makers?

Ans. A Worksheet contains a single visualization. A Dashboard combines multiple worksheets into one interactive view. Dashboards are more useful because decision-makers can see multiple insights in one place.

Q9. What is Tableau Prep, and why is it important in real-world data projects?

What is Tableau Prep, and why is it important in real-world data projects?

Then explain how you would:

Remove null values

Remove duplicate records

Change incorrect data types (Answer in steps, no tool access required.)

Ans.

Tableau Prep is a data preparation tool used to clean and transform data before analysis.

Steps:

- Remove Null Values: Apply filters and exclude null records.
- Remove Duplicate Records: Use cleaning options to identify and remove duplicates.
- Change Data Types: Select the field and change it to the correct data type such as Date, Number, or String.

Q10. A retail manager wants answers to the following:

Which category generates the highest profit?

Which region performs poorly?

How does sales change over time?

a) Mention which charts you would use

b) Identify one key insight that management could derive

c) Explain how this helps in business decision-making

Ans.

a) Charts:

- Highest Profit by Category → Bar Chart
- Poor Performing Region → Bar Chart or Map
- Sales Change Over Time → Line Chart

b) Key Insight: Management can identify the most profitable category and underperforming regions.

c) This helps businesses improve strategy, increase profits, and allocate resources effectively.